Week 8 progressive hints

Use one hint at a time. Return to your own work after each hint and read the next only if you reach a new, specific impasse.

W08-Q01-PA

1. A parameter is a fixed feature of the population. An estimand is the population quantity targeted by the analysis.
2. Write the sample count as the random variable X before sampling and as its realised value $x = 84$ after sampling.

W08-Q01-PB

1. Ask whether the repeated-sampling statement concerns one interval or the long-run behaviour of the construction method.
2. Before data are observed, the interval endpoints are functions of the random sample. After observation, both endpoints and the parameter are fixed in the frequentist interpretation.

W08-Q02-PA

1. From $\mathcal{N}(\mu, 36)$, identify σ and calculate $\sigma/\sqrt{n}$.
2. Centre the interval at $\bar{x}$ and move 1.96 standard errors in each direction. The exactness comes from the stated normal population and known variance.

W08-Q03-PA

1. Calculate the mean, then make a deviation-and-squared-deviation list. Remember that the sample variance denominator is $n - 1$.
2. The unknown variance is replaced by S^2. Under the stated normal sampling condition, standardisation uses a t_{n-1} distribution.
3. Use $s/\sqrt{n} = \sqrt{s^2/n}$ and place the supplied t critical value on both sides of $\bar{x}$.

W08-Q04-PA

1. The observed estimate is $\hat{p}(\mathbf{x}) = x/n$. Check both $n\hat{p}$ and $n(1-\hat{p})$.
2. Use the estimated standard error $\sqrt{\hat{p}(1-\hat{p})/n}$ and the supplied 90% critical value.

W08-Q04-PB

1. The conservative margin is $1.96\sqrt{(1/4)/n}$.
2. Impose the required upper bound on that margin, square both non-negative sides and round the resulting lower bound for n upwards.

W08-Q05-PA

1. The concern "below 50" determines a lower-tailed alternative. Standardise the observed sample mean under the null.
2. The p-value is the lower-tail standard-normal probability at the observed statistic. Compare it with α, or compare the statistic with the supplied lower critical value.

W08-Q05-PB

1. Name the probability model under which the p-value is calculated and the set of results counted as at least as incompatible with the null.

2. A decision rule controls repeated-sampling error probabilities; it does not place a probability distribution on the truth of the fixed null hypothesis.

W08-Q06-PA

1. Because the population variance is unknown, standardise with $S/\sqrt{n}$. Count the degrees of freedom.
2. The alternative is two-sided, so equal-magnitude departures in both tails count against the null.
3. Double the supplied upper-tail probability and compare the result with 0.05. Check that the critical-value decision agrees.

W08-Q06-PB

1. Use the same estimated standard error and the same $t_{0.975,9}$ critical value as in the two-sided test.
2. A two-sided level-α test using the same pivot rejects a candidate null value exactly when that value lies outside the corresponding $100(1-\alpha)\%$ interval.

W08-Q07-PA

1. Under the null, use $\sqrt{p_0(1-p_0)/n}$, not the observed-proportion standard error.
2. Standardise $\hat{p} = 0.58$, double the relevant upper-tail probability and compare the absolute statistic with 1.96.

W08-Q08-PA

1. For $n = 16$, the sampling standard deviation of $\bar{X}$ is $4/\sqrt{16}$. Evaluate the fixed rejection boundary under $\mu = 50$.
2. At $\mu = 53$, a Type II error is the event of not crossing the same rejection boundary. Power is its complement.

W08-Q08-PB

1. To make the test have level 0.05, place c_n exactly $z_{0.95}$ standard errors above 50.
2. Express $\beta(52) = \mathbb{P}_{\mu=52}(\bar{X} \leq c_n)$ as a standard-normal probability. Power at least 0.90 means this Type II error probability is at most 0.10.
3. Substitute the expression for c_n, use $z_{0.10} = -z_{0.90}$, isolate $\sqrt{n}$, and check the two adjacent integers.

W08-Q09-PA

1. For the matched two-sided normal procedures, a candidate null value is rejected exactly when it is outside the realised confidence interval.
2. Apply the containment check separately to 100 and 105. Non-rejection is a statement about insufficient evidence at the chosen level, not a probability assigned to one parameter value.

W08-Q10-PA

1. Use linearity of expectation and independence in the variance of the sample mean.
2. The standard deviation of $\bar{X}$ is the population standard deviation divided by $\sqrt{n}$. Standardise both probability endpoints and invoke the central limit theorem because the population distribution is not stated to be normal.

W08-Q10-PB

1. For a Bernoulli random variable, $\mathbb{E}[X_i] = p$ and $\mathrm{Var}(X_i) = p(1-p)$.
2. Apply the sample-mean expectation and variance formulae. For consistency, consider what happens to the bias and variance as $n \to \infty$, then distinguish the random estimator from its observed numerical estimate.

W08-Q11-PA

1. Differentiate the displayed Lagrangian with respect to x and y. Add primal feasibility, $\lambda \geq 0$ and complementary slackness $\lambda y = 0$.
2. For the interior regime set $\lambda = 0$. For the corner regime set $y = 0$, use the equality constraint and then check the sign of the resulting multiplier.
3. Substitute each regime's solution into the objective to obtain $V(a)$. Because the equality is encoded as $a - x - y = 0$, the envelope derivative is the partial derivative of the Lagrangian with respect to a.